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PALEOMAGNETIC EVIDENCE RELEVANT TO THE ORIGIN OF THE VREDEFORT RING[1]

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ABSTRACT

The time of emplacement of three masses of alkali granite and associated syenites, intruded into the outward-overturned collar of sedimentary rocks is crucial to choosing between a meteorite-impact or terrestrial origin for the Vredefort structure. These intrusives contain pseudotachylite, shatter cones, and microscopic evidence of shock deformation, and their solidification must be preimpact if the structure is an astrobleme. However their field relationships are considered to indicate that they were emplaced after the overturning, and hence that their shock features are unrelated to the mechanism of overturning. Radiometric age measurements have shown that the alkali granites are at most 70 m.y. older than the basic granophyre (1,970 m.y.) of a poststructure ring-dike complex. Relatively little polar wander might be expected in such a short interval. The paleomagnetism of this granophyre, the alkali granites, and other relevant intrusive bodies has been investigated to determine whether the paleomagnetic vector in the alkali granite is similar to, or is overturned with respect to, that in the basic granophyre. Widely scattered results, possibly due to the shock deformation, were obtained from the two innermost alkali-granite plutons. Good data, however, were obtained from the third, outer alkali-granite mass, and from olivine gabbro of the Rietfontein Complex, with which it is closely associated. The magnetic vectors from these two rock masses are about 54 degrees apart, suggesting differential structural rotation. When compared with the vector of the poststructure basic granophyre, however, they are both displaced more than 90 degrees before, and less than 30 degrees after, correction for overturning. It is therefore concluded that the basic igneous complex and the outer alkali granite on the farm Rietfontein, and presumably the other two alkali-granite plutons as well, were intruded prior to the development of the Vredefort Ring structure. The ring must have formed in the 70 m.y. or less between the times of emplacement of the granites and the basic granophyre, and could have been produced by impact.

INTRODUCTION

The general geology of the Vredefort Ring (fig. 1) has been described in detail by Hall and Molengraaff (1925) and Nel (1927). It consists of a central core of basement granite, 26 miles in diameter, surrounded by an upturned to overturned collar of sediments and volcanics totalling 45,000 feet in thickness. To the axis of the surrounding ring syncline, the structure is 60 miles in diameter. The sediments are intruded by younger alkali-granite plutons on the northwest side of the ring, and a basic granophyre ring-dike complex closely follows the outer edge of the granite core (see also Willemse 1937).

The granite core, and the immediately surrounding sediments and volcanics (including the younger alkali granites but excluding the basic granophyre ring dike) are profusely cut by veins of "ultramylonite" or "pseudotachylite" (Bisschoff 1962*a*), and contain abundant shatter cones (Hargraves 1961). The orientation of these cones is such that if the overturned rocks are ro-

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tated back to the horizontal, the core apices point radially inward and upward to a focus approximately 40,000 feet above the granite (Manton 1965). There is widespread microfracturing in Vredefort rocks. Carter (1965) found basal deformation lamellae in quartz from a Vredefort collar shatter cone. These features all provide evidence for an origin by impact (Boon and Albritton 1936; Daly 1947; Dietz 1961).

The principal published arguments against such an origin are these:

1. The younger alkali granite, which also contains shatter cones and pseudotachylite, is considered by those who have mapped it to have been intruded after the overturning of the collar sediments (Nel 1927; see also Bisschoff 1962*a*, 1962*b*). This interpretation is based on the radially outward dip of the sediments on three sides of the largest of the three granite plutons. On this basis the shock deformation cannot be diagnostic of the mechanism of formation of the Vredefort Ring.

2. The basic granophyre ring-dike is clearly postoverturning, but apparently is little younger than the deformation. It should, therefore, date the formation of the structure. However, radiometric age determinations on the basic granophyre (1,970 m.y.) are not significantly different from those on the alkali granite (2,040 m.y., Nicolaysen et al. 1963). Therefore, the granophyre is little, if any, younger than the alkali-granite intrusions, which on the impact hypothesis, must necessarily be "preimpact," and hence completely unrelated to the granophyre.

3. The Vredefort Ring can be viewed as located on a lineament which includes the Great Dyke, the Bushveld Complex, and the Trompsburg Complex (Cousins 1960; Bucher 1963; fig. 2). The Great Dyke (2,530 m.y., Allsopp 1965) and Trompsburg (1,400 m.y., W. I. Manton, personal communication) are now known to be quite different in age, so they do not bear on the problem; but the ages of the Bushveld Complex and of the Vredefort alkali granite and basic granophyre are all within about 70 m.y. (Nicolaysen et al. 1958, 1963). The apparent contemporaneity of the Vredefort Ring and the Bushveld Complex at 1,970 m.y. suggests related mechanisms of formation.

4. The Vredefort Ring is located precisely in the middle of the large elliptical basin of Witwatersrand sedimentary rocks (Brock 1950; Brock and Pretorius 1964; see fig. 2). The geology of the basin, with its remarkable gold and uranium deposits, has been studied in detail. Sedimentologic evidence is alleged by Brock and Pretorius (1964) to indicate the presence of an ancestral Vredefort Dome during the deposition of the Witwatersrand strata, although their conclusion is not unequivocally required by the data.

This paper deals primarily with the substance of the first of these four anti-impact arguments. Notwithstanding the presence of shatter cones and pseudotachylite, if the alkali granite was indeed intruded after the overturning of the Vredefort collar, this fact would provide a specific and devastating argument against the impact theory.

THESIS

The radiometric data imply a difference in age of no more than 70 m.y. between the alkali granite and the poststructure basic granophyre (Nicolaysen et al. 1963). If the alkali-granite masses were intruded after the overturning, their primary remanent magnetism should be similar to that of the basic granophyre, assuming restricted polar wander during this short interval. On the other hand, if the alkali granites were intruded before overturning, they should have been rotated with the collar rocks, through angles ranging up to 130 degrees. If their primary remanence survives the overturning, the hypothesis predicts a specific angular relationship between the vectors of the alkali granite and the basic granophyre.

FIELD AND LABORATORY METHOD

Sampling.—At each of the sites indicated in figure 1, from one to four core-holes (usually three), 60–80 cm deep were drilled, using a portable diamond core-drill (Graham and Keiller 1960). The necessity for collecting samples for paleomagnetic study from well below the surface outcrop because of the frequency of lightning strikes in southern Africa has been demonstrated by Graham (1961). The cores were oriented

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with the aid of a sun compass (La Rochelle 1961). At least two core specimens, 2.0 cm long, from the bottom of each hole, were used for the magnetic measurements.

Magnetic measurements.—Orientation of remanent magnetism was measured on a spinner magnetometer (both an instrument described by Phillips and Kuckes 1967, and a Princeton Applied Research Model SM-1 magnetometer were used in this study). Accuracy of measurements were estimated to be within 5 degrees. Alternating field (A.F.) and thermal demagnetization was performed on standard equipment described by McElhinny (1963) and Phillips (1964). The results obtained in this study are presented in table 1.

ROCK TYPES AND MAGNETIC MEASUREMENTS

1. *Basic granophyre: sites 1, 2, 3, and 4.*—The basic granophyre occurs as a series of vertical or steep dikes in the northwest quadrant of the Vredefort Ring (fig. 1). Some dikes occur isolated in the central granite (e.g., east of Vredefort), whereas others cut both the granite and contiguous Lower Witwatersrand sediments, roughly paralleling the contact between them. The

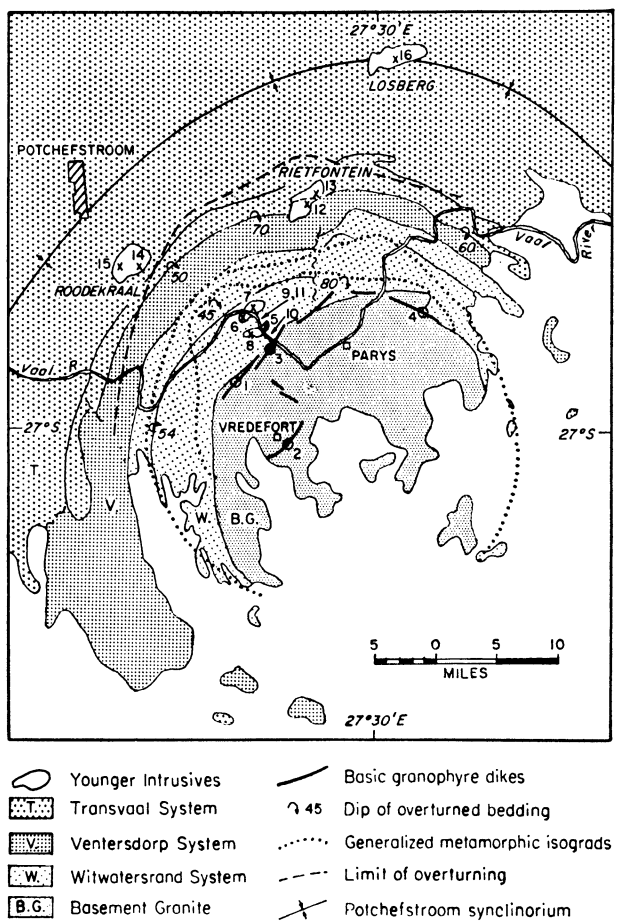


FIG. 1.—Geological map of the Vredefort area, showing location of sites sampled for paleomagnetic study—1–4 (circles) are in basic granophyre; 5–8 (crosses) are in alkali granite of the Vaal River bosses; 13, alkali granite of Rietfontein; other sampling sites as described in table 1. Geology compiled from Manton (1962, pl. *B*) and Bisschoff (1962*a*).

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arcuate distribution is suggestive of a ring dike.

The rock consists of hypersthene and sodic labradorite with interstitial, granophyrically intergrown quartz and potassium feldspar (Willemse 1937). Biotite and very fine-grained titaniferous magnetite are the common accessory minerals. Some composite dikes have spherulitic margins and granular cores. All dikes are characterized by abundant inclusions; these include granite, epidiorite, and hornfels, but are predominantly quartzite, even in dikes in granite over 4 miles from the nearest surface outcrop of Witwatersrand rocks.

With respect to their black, aphanitic chilled contacts and their numerous inclusions, the dikes resemble the pseudotachylite, and may have a common origin (Hall and Molengraaff 1925; Nel 1927). By virtue of their ring-dike form, their resemblance to pseudotachylite, and their lack of deformation, the granophyres are generally thought to be related to the formation of the Vredefort structure, but to immediately postdate it. The basic granophyre is thus considered to date the origin of the structure (1,970 m.y., Nicolaysen et al. 1963).

The natural remanent magnetism (NRM) of the dike samples was widely scattered, but they responded well to A.F. demagnetization, the greatest within- and between-site consistency being obtained after 500 oersteds (see table 1). A paleomagnetic north pole at about 22° N, 27° E ($\alpha = 14$, $\delta = 18$) is indicated.

The basic granophyre is of virtually the same age as the Bushveld granite. The NRM of Bushveld gabbro was studied by Gough and Van Niekerk (1959) before magnetic washing became standard practice.

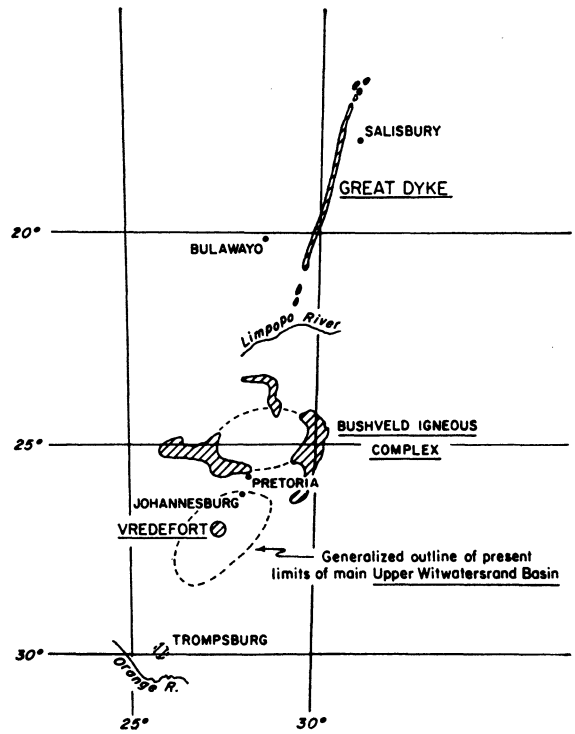


FIG. 2.—Map showing location of the Vredefort Ring in relation to the Upper Witwatersrand Basin, the Great Dyke, the Bushveld igneous complex and the Trompsburg body. Modified after Cousins (1960).

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TABLE 1

PALEOMAGNETIC DATA OBTAINED FROM VREDEFORT SAMPLES, BEFORE AND AFTER DEMAGNETIZATION

Rock Type	Site No.	Samples (N)	Natural						Demagnetized						Sites (N)	D	I	K	α95
			D_n	I_n	K	α95	J	χ	A.F.	D	I	K	α95	J					
Basic granophyre	1	3	345	48	3	88	2.2−3	1.1−3	500	350	65	110	12	1.1−3	4	00	61	62	12
	2	3	351	45	1	....	2.0−3	8.5−4	500	22	61	43	19	2.4−4					
	3	3	319	17	2	....	2.1−3	1.0−3	500	341	61	112	12	8.2−4					
	4	3	33	56	1	....	7.2−4	2.4−4	500	04	52	67	15	3.0−5					
Alkali granite (Vaal River bosses)	5	3	152	−43	12	38	9.7−4	3.5−4	500	127	−55	1	.....	6.8−6	.....	.....	.....	.....	.....
	6	3	321	−17	3	....	8.0−4	2.6−4	500	346	14	7	49	6.9−6	.....	.....	.....	.....	.....
	7	3	124	31	2	....	1.7−3	2.1−4	500	311	22	1	.....	3.0−6	.....	.....	.....	.....	.....
	8	3	318	29	1	....	2.8−4	4.0−5	500	322	10	2	.....	2.9−6	.....	.....	.....	.....	.....
Nepheline syenite	9	3	45	−17	2	....	5.9−3	1.0−3	3,500	32	40	8	48	1.6−5	3	20	47	45	19
Hornfels	10	2	275	−36	3	....	.....	3.3−4	300	01	48	55	.....	4.0−5					
Cordierite hornfels	10	(1)	78	33	30	....	6.8−6	4.5−5	.....	.....	.....	.....	.....	.....					
Dolerite (Group 3)	11	1(2)	318	47	(361)	....	6.4−3	3.2−4	500	24	50	(6,370)	.....	3.0−3					
Rietfontein:																			
Olivine gabbro	12	4	217	21	4	54	3.0−2	3.9−3	500	187	16	120	8	2.3−4	2	158	19	.....	.....
Alkali granite	13	3	330	−12	8	48	5.3−4	5.4−4	5,700	130	18	42	19	1.7−5					
Roodekraal:																			
Porphyritic diorite	14	3	33	53	1	....	3.4−2	2.4−3	5,700	294	58	14	35	1.2−3	.....	.....	.....	.....	.....
Diorite	15	4	78	−31	6	40	2.2−3	6.3−3	700	103	79	14	25	.....	.....	.....	.....	.....	.....
Losberg:																			
Gabbro	16	3	317	30	2	....	1.2−3	4.4−4	500	09	48	182	9	7.2−6	.....	.....	.....	.....	.....

NOTE.—Sites numbered as shown on figure 1. At least two specimens from each sample were measured; D, I refer to declination and inclination of remanence, the subscript n referring to NRM; K and α95 are Fisher's K, and the half-angle of the cone of 95 percent confidence; J = intensity, in electromagnetic units per cubic centimeter, as for example: 2.2−3 means 2.2×10^{-3} emu/gram; χ is susceptibility in electromagnetic units per cubic centimeter oersted; A.F. indicates the intensity of the alternating field used for demagnetization in oersted.

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Stability of the NRM was inferred from the successful application of a fold-test, and gave a north pole at 23 N, 36 E. This result is not significantly different from the pole obtained from the basic granophyre in this study.

2. *Alkali granite and nepheline syenite: sites 5, 6, 7, 8, 9, and 13.*—Three small bosses of granite are intruded into the collar of upturned sediments and volcanics surrounding the central granite core (fig. 1). Two—the Vaal River bosses—are emplaced in the lower Witwatersrand System; the third, very poorly exposed, occurs in the Transvaal Dolomite, close to the contact with the Ventersdorp lavas. This latter granite boss also intrudes, but is closely associated with, an ultrabasic igneous body, the whole known as the Rietfontein complex (Bisschoff 1969).

These bosses are composed of distinctive massive, medium-to-fine-grained, leucocratic, very sodic granite (Molengraaff and Hall 1924; Tilley 1960; Bisschoff 1969). Arfvedsonite and biotite, with or without aegirine, are the major mafic minerals.

The Vaal River bosses are cut by irregular, uneven-grained dikes composed principally of nepheline and albite, with microcline, cancrinite, aegerine, and biotite. The dikes occur also in the country rock between the two bosses, but appear to be genetically related to the granites, and have a similar radiometric age (Nicolaysen et al. 1963).

The granites and the dikes are cut by numerous veins of pseudotachylite, and show conspicuous crushing, shatter cones, and microscopic shock-deformation features.

The magnetic results (see table 1) obtained from the four sites in the Vaal River bosses (sites 5, 6, 7, and 8) remained widely scattered, despite all efforts at A.F. and thermal demagnetization. While adjacent specimens from the same core showed modest agreement, there was considerable scatter between cores from the same site. Only the results from the nepheline syenite (site 9) are significantly consistent.

The Rietfontein alkali granite (site 13) gives good data after A.F. demagnetization. This boss is further from the center of the ring structure, and although veins of pseudotachylite occur, they are much less abundant than in the Vaal River granites. Despite efforts to avoid them, pseudotachylite veins were encountered in every core drilled in alkali granite in the Vaal River bosses. Microdeformation is also much more conspicuous in these granites than at Rietfontein, and the contrast in their remanent magnetic properties is tentatively attributed to these shock effects.

3. *Hornfels: site 10.*—Thermal metamorphism of the Lower Witwatersrand shales has led to a spectacular development of andalusite, cordierite, and garnet hornfels over a wide area in the Vredefort collar, especially on the northwest side (fig. 1). These metamorphosed rocks are also cut by pseudotachylite veins. Although the most intense metamorphism occurs in the vicinity of the Vaal River alkali-granite bosses, these alone are too small to account for the enormous aureole, which has been postulated as due to a large unexposed pluton. A narrow aureole is conspicuous around the Rietfontein granite (Hall and Molengraaff 1925).

On the farm Koedoeslaagte, three cores were drilled (site 10), one each in garnet-grunerite, andalusite, and cordierite hornfels. The intensity of remanence in all three rock types was weak but proved to be very stable with respect to A.F. demagnetization. The remanences of the garnet and andalusite hornfels are very similar and quite distinct from that of the cordierite hornfels, which is near that of the present earth's field (see table 1). Only the results from the garnet and andalusite hornfels samples have been averaged and used in the subsequent analysis.

4. *Other basic intrusives: site 11.*—Bisschoff (1962*b*) has distinguished four distinct stages of doleritic intrusive activity in the Vredefort Dome. The oldest (Group 1) is older than the basement granite of the central core, whereas the other three are all intrusive into the collar sediments. The

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youngest, Group 4, constitutes horizontal sheets transgressing the overturned strata, and are regarded as of Mesozoic (late Karroo) age. Dolerites of Groups 2 and 3 are more or less conformable with the sediments, are cut by pseudotachylite, and "display the features of inverted sills" (Bisschoff 1962*b*, p. 249; 1969, p. 7), and hence are considered to have been emplaced prior to the overturning.

One core was drilled from a small isolated outcrop of a dolerite of Group 3 (site 11). Separate specimens from this core gave identical results, which were extremely stable with respect to A.F. demagnetization.

5. *Rietfontein complex: sites 12 and 13.*—This complex is (Bisschoff 1962*a*, 1962*b*, 1969) intruded into the Transvaal Dolomite, and is described as a strongly differentiated mass, consisting in its southern half of sodic olivine gabbro, wehrlite, and troctolite, cut by alkali granite composing the northern half. Bisschoff regarded the structure of the ultrabasic part of the complex as a discordant funnel-shaped intrusion. The country rock is overturned, but the distribution of rock types indicates the complex itself to be right side up; thus, the intrusion postdates the Vredefort deformation.

Four cores were drilled in an outcrop of olivine-rich gabbro (site 12), and three in a nearby outcrop of the alkali granite (site 13). The gabbro is fresh, showing only slight serpentinization of the abundant olivine. A vague suggestion of layering visible in this outcrop dips 60° toward 190°.

The internal consistency of the paleomagnetic data after A.F. demagnetization at 500 oersteds is excellent in the gabbro and good in the granite. The average remanent vectors from these two sites are, however, about 54 degrees apart (table 1).

6. *Roodekraal complex: sites 14 and 15.*—This intrusion (fig. 1) occupies an area of 5 square miles and consists of highly varied pyroxene diorite, hornblende quartz diorite, and andesite, with feldspars ranging from An0 to An35 (Bisschoff 1962*b*, p. 250; Nel et al. 1939, p. 90). A common phase is of diorite rock with conspicuous plagioclase phenocrysts up to an inch in length (site 14). An equigranular, medium-to-fine-grained phase was sampled at site 15. Some phases with conspicuous vesicles appear to be volcanic and may be extrusive andesite (Bisschoff 1962*a*, p. 218). Hydrothermal alteration of both feldspars and ferromagnesians is ubiquitous at both sites sampled. Bisschoff (1962*a*, p. 218) inferred the emplacement of the complex to postdate the diastrophism of the Vredefort Dome and the Potchefstroom synclinorium (Nel et al. 1939; the ring-syncline of fig. 1).

The intensity of remanence in the three cores drilled in porphyritic diorite (site 14), decreases rapidly with A.F. demagnetization, but moderate consistency in orientation is achieved after 500 oersteds (table 1).

Four medium-to-fine-grained equigranular diorite samples from site 15 (oriented hand samples collected in a quarry) have a mean NRM nearly opposite that of site 14 after demagnetization. With increasing A.F. demagnetization up to 1,000 oersteds, this average slowly changes toward, but never reaches, that of site 14. The greatest within-site consistency is attained after 700 oersteds. The difference between the two sites may be due to differential tectonic rotation, but unfortunately, details of the present structural attitude and possible structural history are not available.

7. *Losberg intrusion: site 16.*—This complex is a flat-lying, differentiated layered body, approximately 400 feet thick, intruded into shales and quartzites of the Transvaal system (Abbott and Ferguson 1965). It is situated on the axis of the syncline marking the outer border of the Vredefort Ring (Potchefstroom synclinorium). Ultrabasic accumulates at the base grade upward into gabbro.

Three cores drilled at one site (site 16) in the gabbro gave highly stable and consistent paleomagnetic data (table 1), virtually identical with that obtained from the basic granophyre and the Bushveld gabbro. These data suggest a Bushveld age for the intrusion, inferred on other grounds by

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Abbott and Ferguson (1965, p. 33), and by Cousins (1960).

SUMMARY AND DISCUSSION

The remanent magnetism vectors for the rocks measured in this study are compared with those equivalent to the polar-wander curve for southern Africa (McElhinny et al. 1968) in figure 3. Samples from four sites in

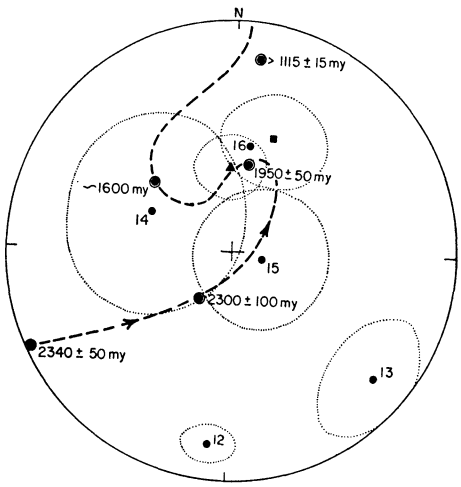


FIG. 3.—Equal-area lower-hemisphere projection summarizing paleomagnetic data in table 1. The heavy broken line linking double solid circles portrays the migration of the earth's magnetic field vector at Vredefort, according to the polar-wander curve of McElhinny et al. 1968. Solid triangle—basic granophyre; solid square—mean of sites 9, 10, 11; results from other sites numbered accordingly. The 95 percent confidence cone for site 16 (9°) has been omitted for clarity. Solid circles are of north-seeking pole on the lower hemisphere; present-day north-seeking pole would be on upper hemisphere.

the posttectonic basic granophyre ring-dike complex give consistent, stable paleomagnetic data, and their mean remanent vector is almost identical with that of the Bushveld gabbro which is of the same age (1,970 m.y.).

With respect to the younger igneous complexes or plutons sampled (12 sites), whose time of intrusion relative to Vredefort diastrophism is in question:

1. The Losberg intrusion (site 16) gives consistent paleomagnetic data, of the same orientation as that from the basic granophyre. This complex is emplaced on the axis of the Potchefstroom synclinorium, however; since it and its conformable country rock are almost flat-lying, the result cannot discriminate between emplacement pre- or post-Vredefort diastrophism.

2. The results from Roodekraal (sites 14 and 15) are somewhat scattered; the site mean vectors are 43 degrees apart, but both are within 35 degrees of the basic granophyre vector (see table 1 and fig. 3). Because of the large within-site dispersions, however, none of these angular differences can be considered significant. Furthermore, the structural environment around Roodekraal is complex, and the attitude of the intrusion so uncertain that no fold-test can be made with confidence.

3. Of the Vaal River sites (all in overturned country rock), the widely scattered results from the alkali-granite bosses are tentatively attributed to shock metamorphism.

The results from the nepheline syenite (site 9, $\alpha 95 = 48°$), hornfels (site 10, only two samples utilized), and Group 3 dolerite (site 11, one sample only) are of moderate and uneven quality. Their mean vectors are all quite similar however, and for convenience of discussion they have been grouped and averaged (table 1). This average vector is similar to that of the basic granophyre (fig. 3), and it would seem that either the metamorphism and igneous intrusion which these rocks represent is poststructure, or they have all undergone poststructure remagnetization, more or less contemporaneous with the emplacement of the basic granophyre.

The former alternative (poststructure emplacement) is, by itself, the most plausible. The possibility of poststructure remagnetization (second alternative) however, is indicated by the intense development of pseudotachylite in this vicinity. All the above rocks are cut by veins of pseudotachylite and Bisschoff (1962*a*) reports their recrystallization equivalent to amphibolite facies conditions. A thermal event of this

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magnitude (>500° C?) even if relatively transient, would be sufficient to cause remagnetization of the affected rocks.

This interpretation is favored, as it is the only one in keeping with the results from Rietfontein. However, it is acknowledged that these results alone are ambiguous.

4. The two Rietfontein sites both gave very good mean paleomagnetic vectors (site 12, $\alpha 95 = 8°$, site 13, $\alpha 95 = 19°$); although they are 54 degrees apart, they are quite different from any others of similar age, or from the present earth's field (fig. 3).

The Rietfontein igneous rocks sampled are intruded into Transvaal sediments, which are now overturned by at least 20 degrees. The remanence reported is highly stable, and as the rocks show only rare pseudotachylite veins and negligible microscopic evidence of metamorphism or alteration, there is little reason to suspect secondary magnetization. In fact the 54-degree difference between the two vectors (sites 12 and 13) precludes the possibility of posttectonic remagnetization as is postulated to have occurred along the Vaal River. This angular discrepancy between the two sites could, by itself, be considered to suggest a significant difference in age for the two rock types if they were intruded after the overturning, or if the possibility of differential tectonic rotation is neglected (see Bisschoff 1962*b*, p. 250).

On the other hand, if a preoverturning emplacement for both is contemplated, then differential rotation is both conceivable, and has been recorded in Witwatersrand sediments in the immediate vicinity. By comparing cleavage-bedding intersection lineations with the orientation of shattercone axes, Manton (1965, p. 1038) showed that there had been differential rotations of large blocks of up to 30 degrees about axes perpendicular to bedding. As the vectorial differences would be accentuated by slight differences in the axis of overturning, the discrepancy in paleomagnetic orientation between the two sites is not unreasonable.

The contrast between these paleomagnetic vectors and that from the basic granophyre is considerable. With respect to the original test proposed, according to Geological Survey maps, the general attitude of the Transvaal system in the vicinity of Rietfontein is strike, N 75° E; overturned dip, 70° SE. As may be seen in figure 4, the Rietfontein vectors agree quite well (within 30 degrees) with that of the basic granophyre after rotating 110 degrees to correct for the overturned bedding.

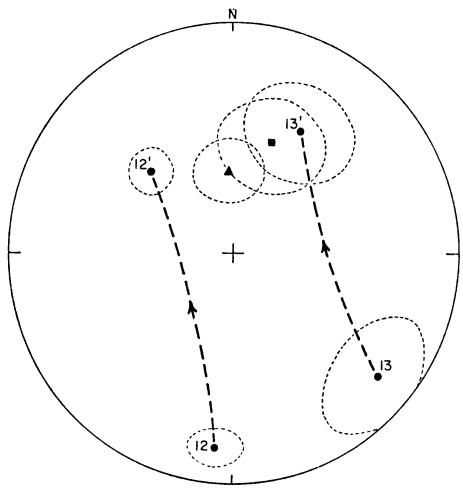


FIG. 4.—Equal-area lower-hemisphere projection showing position of basic granophyre vector (solid triangle) and mean of sites 9, 10, and 11 (solid square) for comparison with those from Rietfontein before (12 and 13) and after (12′, 13′) restoration to horizontal of the Rietfontein country rock.

It seems unlikely that this agreement can be fortuitous, and the result is considered to confirm the hypothesis that the Rietfontein basic complex and its associated alkali-granite pluton (and presumably the Vaal River plutons as well) were intruded into essentially flat-lying sediments, prior to the development of the Vredefort Ring structure. This result does not prove the meteorite impact hypothesis. However, it does demonstrate that the entire structural development must have occurred in the restricted interval subsequent to the emplacement of the alkali granite, and prior to the crystallization of the basic granophyre.

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REFERENCES CITED

ABBOTT, D., and FERGUSON, J., 1965, The Losberg intrusion, Fochville, Transvaal: Geol. Soc. South Africa, Trans., v. 68, p. 31–52.

ALLSOPP, H. L., 1965, Rb-Sr and K-Ar measurements on the Great Dyke of southern Rhodesia: Jour. Geophys. Research, v. 70, p. 977–984.

BISSCHOFF, A. A., 1962*a*, The pseudo-tachylite of the Vredefort Dome: Geol. Soc. South Africa, Trans., v. 65, p. 207–225.

——— 1962*b*, Contribution to discussion of paper "Notes on the Vredefort Dome," by A. Poldervaart: Geol. Soc. South Africa, Trans., v. 65, p. 249–251.

——— 1969, The petrology of the igneous and the metamorphic rocks in the Vredefort Dome and the adjoining parts of the Potchefstroom syncline: D.Sc. thesis, University of Pretoria, South Africa, 243 p.

BOON, J. D., and ALBRITTON, C. C., 1936, Meteorite craters and their possible relationship to cryptovolcanic structures: Field and Lab., v. 5, p. 1–9.

BROCK, B. B., 1950, The Vredefort Ring: Geol. Soc. South Africa, Trans., v. 53, p. 131–144.

———, and Pretorius, D. A., 1964, Rand basin sedimentation and tectonics, *in* The geology of some ore deposits of southern Africa: Geol. Soc. South Africa, v. 1, p. 549–599.

BUCHER, W. H., 1963, Cryptoexplosion structures caused from without or from within the earth? ("Astroblemes" or "Geoblemes"?): Am. Jour. Sci., v. 261, p. 597–649.

CARTER, N. L., 1965, Basal quartz deformation lamellae—a criterion for the recognition of impactites: Am. Jour. Sci., v. 263, p. 786–806.

COUSINS, C. A., 1960, The structure of the mafic portion of the Bushveld igneous complex: Geol. Soc. South Africa, Trans., v. 62, p. 179–201.

DALY, R. A., 1947, The Vredefort ring structure of South Africa: Jour. Geology, v. 55, p. 125–145.

DIETZ, R. S., 1961, Vredefort ring structure: meteorite impact scar?: Jour. Geology, v. 69, p. 499–516.

GOUGH, D. I., and VAN NIEKERK, C. B., 1959, A study of the paleomagnetism of the Bushveld gabbro: Philos. Mag., v. 4, p. 126–136.

GRAHAM, K. W., 1961, The re-magnetization of a surface outcrop by lightning currents: Geophys. Jour. v. 6, p. 85–102.

———, and KEILLER, J. A., 1960, A portable drill rig for producing short oriented cores: Geol. Soc. South Africa, Trans., v. 63, p. 71–74.

HALL, A. L., and MOLENGRAAFF, G. A. F., 1925, The Vredefort mountain land in the southern Transvaal and northern Orange Free State: Shaler Memorial Series, Koninkl. Akad. Wetensch. Amsterdam, Verh. 24, 183 p.

HARGRAVES, R. B., 1961, Shatter cones in the rocks of the Vredefort Ring: Geol. Soc. South Africa, Trans., v. 64, p. 147–161.

LA ROCHELLE, A., 1961, Adaptation of an astrocompass for collection of oriented rock specimens: Geol. Survey Canada. Topical Rept., no. 32, 4 p.

MANTON, W. I., 1962, The orientation and implication of shatter cones in the Vredefort Ring structure: M.Sc. thesis, Univ. Witwatersrand, 167 p.

——— 1965, The orientation and origin of shatter cones in the Vredefort Ring: Annals New York Acad. Sci., v. 123, p. 1017–1049.

MCELHINNY, M. W., 1963, Theory and operating procedures: rock magnetism instruments at Princeton University: Princeton Univ. Dept. Geol. Eng. Rept., no. 63-1, 24 p.

———; BRIDEN, J. C.; JONES, D. L.; and BROCK, A., 1968, Geological and geophysical implications of paleomagnetic results from Africa: Rev. Geophys., v. 6, p. 201–238.

MOLENGRAAFF, G. A., and HALL, A. L., 1924, Alkali granite and nepheline syenites, canadite and foyaite, in the Vredefort mountain land, South Africa: Koninkl. Akad. Wetensch. Amsterdam, Proc. 27, p. 465–486.

NEL, L. T., 1927, The geology of the country around Vredefort: Geol. Survey South Africa, Spec. Pub. no. 6, 134 p.

———; TRUTER, F. C.; WILLEMSE, J.; and MELLOR, E. T., 1939, The geology of the country around Potchefstroom: Geol. Survey South Africa, Expl. Sheet 61., 156 p.

NICOLAYSEN, L. O., DE VILLIERS, J. W. L.; BURGER, A. J.; and STRELOW, F. W. E., 1958, New measurements relating to the absolute age of the Transvaal system, and of the Bushveld igneous complex: Geol. Soc. South Africa, Trans., v. 61, p. 137–163.

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———; BURGER, A. J.; and VAN NIEKERK, C. B., 1963, The origin of the Vredefort Dome structure in the light of new isotopic data (abs.): Internat. Union Geodesy and Geophysics, 13th General Assembly, Berkeley, Calif.

PHILLIPS, J. D., 1964, A thermal demagnetisation device: Princeton Univ., Dept. Geol. Eng. Rept. no. 64-3, 25 p.

———, and KUCKES, A. F., 1967, A spinner magnetometer: Jour. Geophys. Research, v. 72, p. 2209–2214.

TILLEY, C. E., 1960, Some new chemical data on the alkali rocks of the Vredefort mountain land, South Africa: Geol. Soc. South Africa, Trans., v. 63, p. 65–70.

WILLEMSE, J., 1937, On the old Granite of the Vredefort region and some of its associated rocks: Geol. Soc. South Africa, Trans., v. 40, p. 43–120.

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